Homework: Gardener Problem

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	[,1]	[,2]	[,3]
[1,]	0.3	0.6	0.1
[2,]	0.1	0.6	0.3
[3,]	0.05	0.4	0.55

Consider the gardener problem with fertilizer. Suppose that the cost of the fertilizer is \$50 per bag and the garden needs two bags if the soil is good. The amount of fertilizer is increased by 25% if the soil is fair and 60% if the soil is poor.

The gardener estimates that annual yield to be worth \$250 if no fertilizer is used and \$420 if fertilizer is applied. Is it worthwhile to use the fertilizer?

- Write balance equations to determine the steady state probabilities π_1, π_2, π_3
- Determine steady state probability of good soil
- Suppose fertilizer is applied.
  - Determine expected total cost.
  - Should fertilizer be applied?
- (Optional*) Determine the mean first return time to good soil.

State k	1: good	2: fair	3: poor
Cost of fertilizer $c(k)$	100	125	160

balance equation

$\Pi_1 = 0.3\Pi_1 + 0.1\Pi_2 + 0.05\Pi_3$

$\Pi_2 = 0.6\Pi_1 + 0.6\Pi_2 + 0.4\Pi_3$

$\Pi_3 = 0.1\Pi_1 + 0.3\Pi_2 + 0.55\Pi_3$

normalization equation

$\Pi_1 + \Pi_2 + \Pi_3 = 1$

Steady state probability

$\Pi_1 = 0.102$ (good)

$\Pi_2 = 0.525$ (fair)

$\Pi_3 = 0.373$ (poor)

expected total cost

$= 100(\Pi_1) + 125(\Pi_2) + 160(\Pi_3)$

$= 10.2 + 65.625 + 59.68$

$= 135.5$

profit

- fertilizer
  $= 420 - 135.5$
  $= 284.5$
- no fertilizer
  $= 250$

∴ Should applied fertilizer because get more profit.

first return time

$M_1 = \frac{1}{\Pi_1} = \frac{1}{0.102}$

$M_1 = 9.8$

Homework: Bank inactivity collection fee

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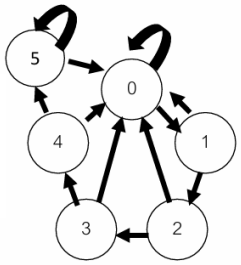
- Consider the steady state.
- Write down the steady-state balance equations.
- Let the probability that account is inactive next month be $r = 0.30$. Solve for the steady-state distribution.
- Recall that the fee is charged each month while the account remains in state 5.
  Let F be the monthly fee. Then, the long-run expected fee revenue per account per month is

$$\mathbb{E}[fee] = F\,\pi_5 \qquad \text{(THB/account/month)}$$

  Compute the expected fee for F= 75 THB.
- (Optional) Derive a closed-form expression for $\pi_5(r)$.

State	Description
0	Active this month
1	1 month inactive
2	2 months inactive
3	3 months inactive
4	4 months inactive
5	More than 4 months inactive (5 months or more inactive)

$$P = \begin{bmatrix} 1-r & r & 0 & 0 & 0 & 0 \\ 1-r & 0 & r & 0 & 0 & 0 \\ 1-r & 0 & 0 & r & 0 & 0 \\ 1-r & 0 & 0 & 0 & r & 0 \\ 1-r & 0 & 0 & 0 & 0 & r \\ 1-r & 0 & 0 & 0 & 0 & r \end{bmatrix}$$



balance equation

$\Pi_0 = 0.7\Pi_0 + 0.3\Pi_1$

$\Pi_1 = 0.7\Pi_0 + 0.3\Pi_2$

$\Pi_2 = 0.7\Pi_0 + 0.3\Pi_3$

$\Pi_3 = 0.7\Pi_0 + 0.3\Pi_4$

$\Pi_4 = 0.7\Pi_0 + 0.3\Pi_5$

$\Pi_5 = 0.7\Pi_0 + 0.3\Pi_5$

normalization equation

$\Pi_0 + \Pi_1 + \Pi_2 + \Pi_3 + \Pi_4 + \Pi_5 = 1$

steady state probability

$\Pi_0 = \Pi_1 = \dots = \Pi_5 = 1/6$

expected fee

$E[Fee] = F \cdot \Pi_5$

$= (75)(\frac{1}{6})$

$E[Fee] = 12.5$

derive closed-form $\Pi_5(r)$

$\Pi_5(r) = \frac{1}{6}$